

C++ Final Post Training Assessment

Test Reference No: PLMAcademy-Apr2026-General-Batch

Total Marks : 30

Test Duration : 1 hour

Instructions to Candidate

1. All questions are compulsory/mandatory.
2. Do not write anything on the question paper.
Write everything on the answer sheet provided.
3. This is a closed book test.
4. There is no negative marking.
5. Wherever there is ambiguity, make suitable assumptions and proceed.
Document the assumptions in the answer.
6. Assume C++ 11 language standard.
7. Assume usage of Microsoft Visual Studio development platform.

Question No.	Question	Marks
1	Will the following code compile and execute successfully? If so, give output or error accordingly. <pre>#include <iostream> using namespace std; template<typename T> void func(T x) { cout << "Template\n"; } template<> void func<int>(int x) { cout << "Specialized\n"; } void func(int x) { cout << "Overloaded\n"; } int main() { func(10); }</pre>	1
2	Explain Dynamic casting with minimal code.	2
3	Will the following code compile and execute successfully? If so, give output or error accordingly. <pre>#include <iostream></pre>	1

	<pre>using namespace std; void fun1() {} void fun2() {} int main() { if (fun1 == fun2) cout << "Same"; else cout << "Different"; }</pre>	
4	Will the following code compile and execute successfully? If so, give output or error accordingly. <pre>#include <iostream> using namespace std; template<typename T = int> void func(T x) { cout << x << endl; } int main() { func<>(10.5); }</pre>	1
5	Will the following code compile and execute successfully? If so, give output or error accordingly. <pre>#include <iostream> using namespace std; int main() { try { int x = 10; throw &x; } catch (int x) { cout << "Value\n"; } catch (int* p) { cout << "Pointer\n"; } }</pre>	1
6	Explain Shared Pointer with minimal code.	2
7	Will the following code compile and execute successfully? If so, give output or error accordingly.	1

	<pre> #include <iostream> using namespace std; int main() { try { throw 'a'; } catch (int x) { cout << "int\n"; } catch (char x) { cout << "char\n"; } catch (...) { cout << "default\n"; } } </pre>	
8	What will be the output of the code below? <pre> #include <iostream> using namespace std; class Base { public: virtual ~Base() {} }; class Derived : public Base {}; int main() { Base b; try { Derived& d = dynamic_cast<Derived&>(b); } catch (bad_cast& e) { cout << "Caught\n"; } } </pre>	1
9	Explain Virtual Destructor with minimal code.	2
10	Write a program that returns the elements in a vector that are smaller than their adjacent left and right neighbors. Input-> 1 2 5 0 3 1 7 Output-> 0 1	2
11	Can the constructor be made virtual in C++? For e.g., what	1

	will be the output of the following code? <pre>#include <iostream> using namespace std ; class X { public: virtual X() { cout << "X constructor called" << endl ; } }; int main () { X x1 ; return 0 ; }</pre>	
12	Why should a function not return the address of a local non-static variable? Explain with minimal code and memory diagram.	2
13	Explain Singleton Design pattern with proper code.	2
14	What will be the output of the following code? <pre>#include <iostream> using namespace std; int main() { shared_ptr<int> p1 = make_shared<int>(50); weak_ptr<int> w = p1; cout << w.use_count() << endl; p1.reset(); cout << w.expired() << endl; }</pre>	1
15	What will be the output of the following program: <pre>#include <iostream> using namespace std; class Shape { int x; public: Shape(int y):x(y){}</pre>	1

	<pre> Shape(const Shape& s) { cout << "Shape Copy constructor called"; x = s.x; } }; int main() { Shape x(10), z(20); Shape y = x; z = y; return 0; } </pre>	
16	Write a program to merge 2 sorted linked list.	2
17	Based on the code below, answer the following: 1. How many VTable will be created for the below code? 2. How many VPTR are created for the below code? 3. What is the output of the below code? <pre> #include <iostream> using namespace std; #pragma pack(1) class Shape { int x; public: virtual void f(){} void g(){} }; class Circle : public Shape{ int y; public: void f(){} }; int main() </pre>	3

	<pre> { Shape* s = new Circle(); cout << "Size of Shape class " << sizeof(Shape) << endl; cout << "Size of Circle class " << sizeof(Circle) << endl; return 0; } </pre>	
18	Write code, which would print the elements of an array, which are at even indexes, in the following code. <pre> int main () { int a[] = { 10, 20, 90, 1, 4, 8, 9, 100 }; return 0 ; } </pre>	1
19	What is the output of the following program? <pre> #include <iostream> using namespace std ; int f() { static int i = 100 ; i++ ; return i ; } int main() { cout << f() << endl ; cout << f() << endl ; return 0 ; } </pre>	1
20	What will be the output of the following code? Explain if there is any issue in this code. <pre> struct A; struct B; struct A { shared_ptr b; ~A() { cout << "A destroyed\n"; } }; struct B { </pre>	2

	<pre>shared_ptr<A> a; ~B() { cout << "B destroyed\n"; } }; int main() { shared_ptr<A> pA = make_shared<A>(); shared_ptr pB = make_shared(); pA->b = pB; pB->a = pA; }</pre>	
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